

Óscar C. Vásquez

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Education

Université Pierre et Marie Curie, France

PhD. in Computer Science

2014

University of Santiago, Chile

M Sc. Eng. mention in Industrial Engineering

2010

École Polytechnique, France

M Sc. & Tech. in Applied Mathematics

2008

Experience (in the last 5 years)

Gobierno de Chile-Agencia Nacional de Investigación y Desarrollo (ANID)

Member, Evaluation Group FONDECYT - Engineering 2 (Industrial, Electrical, Informatics, Transports, similars).

2022-

Gobierno de Chile-Comisión Nacional de Acreditación (CNA-Chile)

Peer reviewer, Evaluation for Master and Doctorate (PhD) programs into Earth and Engineering Sciences Committee

2020-

Universidad de Santiago de Chile, Industrial Engineering Department

Full Professor

2024-

Program for the Development of Sustainable Production Systems (PDSPS), Universidad de Santiago de Chile

Director

2022-

PhD. & MSc. in Eng. mention in Industrial Engineering, Universidad de Santiago de Chile

Director

2014-

Gobierno de Chile, Agencia Nacional de Investigación y Desarrollo (ANID), Comité de Evaluación y Orientación de la Cooperación Científica con Chile del Gobierno de Francia (ECOS)

Principal. ECOS230013, Stochastic programming approach for planning activities with remanufacturing under extended producer responsibility in a sustainability setting

2024-2026

Gobierno de Chile, Agencia Nacional de Investigación y Desarrollo (ANID)

Principal. Regular Grant FONDECYT No1211640, Scheduling problem under explorable uncertainty: Theory, Experiments and Practice

2021-2025

CNRS/France, ANID/Chile, CAPES/Brazil.

Assistant. STIC-AmSud N° 19-STIC-02, C2S2 - Constructing Cryptographically Secure Structures

2019-2020

Programa Iberoamericano de Ciencia y Tecnología para el Desarrollo (CYTED)

Principal. Project N°P318RT0165 Red Iberoamericana Industria 4.0

2019-2022

Publications indexed in WoS Core Collection (in the last 5 years)

Natalia Jorquera-Bravo and Óscar C. Vásquez. On the local dominance properties in single machine scheduling problems. *Annals of Operations Research* 338: 335-345. 2024.

Lorena Espinoza Pérez, Andrea Espinoza Pérez and Óscar C. Vásquez. Evaluation of mitigation initiatives for

simultaneous air pollution and municipal solid waste systems: A System Dynamics Approach *Socio-Economic Planning Sciences* 95: 102010. 2024.

Lorena Espinoza Pérez, Andrea Espinoza Pérez and Óscar C. Vásquez. Life cycle assessment of alternatives for industrial textile recycling *Science of the Total Environment* 927: 172161. 2024.

Nélson Troncoso Héctor Cancela, Pedro Piñeyro, Franco Quezada and Óscar C. Vásquez. The product-mold-machine manufacturing problem: complexity, MILP models and constructive heuristics *Computers & Industrial Engineering* 189: 109937. 2024.

Nicolás Valenzuela-Levi, Andrea Teresa Espinoza-Pérez and Óscar C. Vásquez. Market concentration in industrial and commercial non-hazardous waste valorisation: the case of Chile *Waste Management* 177: 106-114. 2024.

Luis Rojo-González, Jocelyn Dunstan, Cristóbal Cuadrado, Denisse Ávalos, Javier Moraga-Correa; Nelson Troncoso and Óscar C. Vásquez. The problem of estimation and forecasting of obesity prevalence using sparsely collected data *Engineering Applications of Artificial Intelligence* 132: 107860. 2024.

Javiera García-Leal, Andrea Teresa Espinoza-Pérez and Óscar C. Vásquez. Sustainable diets optimal design for the massive food services: Economic versus Environmental aspects *The International Journal of Life Cycle Assessment*. 28: 1481-1494. 2023.

Javiera García-Leal, Andrea Teresa Espinoza-Pérez and Óscar C. Vásquez. Towards the sustainable massive food services: An optimization approach *Socio Economic Planning Sciences* 87(A): 101554. 2023.

Valeria D. Muñoz-Soto, Sebastián Dávila, Andrea Teresa Espinoza-Pérez, Luis Rojo-González, Nicolás Valenzuela-Levi and Óscar C. Vásquez. COVID-19, waste production and municipal recycling programs: Insights from Chile to the Global South *Science of the Total Environment* 899: 165388. 2023.

Natalia Jorquera-Bravo, Andrea Teresa Espinoza-Pérez and Óscar C. Vásquez. A multi-objective solution approach for the design of a sustainable and robust system of wastewater treatment plants: The case of Chile *Computers & Industrial Engineering* 179: 109192. 2023.

Andrea Teresa Espinoza-Pérez, Natalia Jorquera-Bravo and Óscar C. Vásquez. Toward a sustainable system of wastewater treatment plants in Chile: A multi-objective optimization approach *Annals of Operations Research* 311: 731-747. 2022.

Lorena Espinoza Pérez, Andrea Espinoza Pérez and Óscar C. Vásquez. Exploring an alternative to the Chilean textile waste: A carbon footprint assessment of a textile recycling process *Science of the Total Environment* 830: 154542. 2022.

Javiera García Leal, Andrea Espinoza Pérez and Óscar C. Vásquez. Corrigendum: A review of the use of linear programming to optimize diets, nutritiously, economically and environmentally. *Frontiers in Nutrition* 9: 850033. 2022.

Fanny Dufossé, Christoph Dürr, Noël Nadal, Denis Trystram and Óscar C. Vásquez. Scheduling with a processing time oracle. *Applied Mathematical Modelling* 104:701-720. 2022.

Andrea Espinoza Pérez, Mariano Frutos, Daniel Rossit, Martín Safe, Fernando Tohmé and Óscar C. Vásquez. Mass customized/personalized manufacturing in industry 4.0 and blockchain: Research challenges, main problems, and the design of an information architecture. *Information Fusion* 79: 44-57. 2022.

Patricio J. Araya-Córdova, Sebastián Dávila, Nicolás Valenzuela-Levi and Óscar C. Vásquez. Income inequality and efficient resources allocation policy for the adoption of a recycling program by municipalities in developing countries: The case of Chile *Journal of Cleaner Production* 309: 127305. 2021.

Patricio J. Araya-Córdova, Sebastián Dávila, Nicolás Valenzuela-Levi and Óscar C. Vásquez. Promoting adoption of recycling by municipalities in developing countries: Increasing or redistributing existing resources? *Resources, Conservation & Recycling* 164: 105173. 2021.

Mariano Frutos, Daniel Rossit, Martín Safe, Fernando Tohmé and Óscar C. Vásquez. A combinatorial analysis of the permutation and non-permutation flow shop scheduling problem *European Journal of Operational Research* 289(3): 841-854. 2021.

Javier Moraga-Correa, Franco Quezada, Luis Rojo-González and Óscar C. Vásquez. *The balanced dispatching problem in passengers transport services on demand* Expert Systems with Applications 177: 114918. 2021

Lorena Espinoza-Pérez, Andrea Teresa Espinoza-Pérez, Óscar C. Vásquez, Ian Vázquez-Rowe and K. Ziegler-Rodríguez. *Closing the gap in the municipal solid waste management between metropolitan and regional cities from developing countries: A life cycle assessment approach* Waste Management 124: 314-324. 2021

Mariano Frutos, Daniel Rossit, Martín Safe, Fernando Tohmé and Óscar C. Vásquez. *Critical paths of non-permutation and permutation flow shop scheduling problems* International Journal of Industrial Engineering Computations 11: 281-298. 2020.

Awards/Scholarships

Premio ICONO 24, Académico Destacado en Internacionalización

Facultad de Ingeniería, UNiversidad d Santiago de Chile

Escala Docente Grant'23

Asociación de Universidades Grupo Montevideo (AUGM)

Escala Docente Grant'19

Asociación de Universidades Grupo Montevideo (AUGM)

Beca Olga Ulianova'19

Universidad de Santiago de Chile

Escala Docente Grant'18

Asociación de Universidades Grupo Montevideo (AUGM)

Movilidad Docente Grant'18

Alianza Pacífico Chile-Colombia-México-Perú

Escala Docente Grant'17

Asociación de Universidades Grupo Montevideo (AUGM)

Escala Docente Grant'16

Asociación de Universidades Grupo Montevideo (AUGM)

Santander Ibero-American Grants. Young Teachers and Researchers'15

Banco Santander

ELAVIO'14 Scholarship

IFORS, International Federation of OR Societies

WINE'13 Scholarship

WINE, Committee and Sponsors (Facebook-Microsoft-Google)

ELAVIO'13 Scholarship

EURO, Asoc. of European OR Societies

EVIC'12 Scholarship

IEEE, Computational Intelligence Society

PhD Foreign Program Scholarship

Government of Chile- National Commission of Research and Technology (CONICYT)

Master Scholarhip

Universidad de Santiago de Chile

Exonération totale d'excellence- Master Scholarship

École Polytechnique

Master Foreign Program Scholarship

Government of Chile- National Commission of Research and Technology (CONICYT)

Best student national award'06

Chilean Engineering College

Students-Postdocs

27. Edmundo Barrientos Palma M Sc. Eng. mention in Industrial Engineering (2023)
26. Fernanda Suazo Morales. M Sc. Eng. mention in Industrial Engineering (2023) (work conjoint with H. López-Ospina)
25. Valentina Ugalde Calderón M Sc. Eng. mention in Industrial Engineering (2023) (work conjoint with S. Dávila-Gálvez)
24. Cristóbal Mauricio Fuentealba, M Sc. Eng. mention in Industrial Engineering (2023)
23. Paloma Nuñez Muñoz. M Sc. Eng. mention in Industrial Engineering (2022)
22. Bárbara Bezares Bobadilla. M. in Integrated Management Systems (2022)
21. Kevin Saldías Lara. M Sc. Eng. mention in Industrial Engineering (2021) (work conjoint with M.T. Santander)
20. Diego Machado Vásquez. M Sc. Eng. mention in Industrial Engineering (2021) (work conjoint with F. Jara)
19. Tomás Torres Escaff. M Sc. Eng. mention in Industrial Engineering (2021) (work conjoint with G. Lagos)
18. César Karque Ortega. M. in Integrated Management Systems (2021)
17. Makarennna Pimentel Copa. M. in Integrated Management Systems (2021)
16. John Marín Bermúdez. M. in Integrated Management Systems (2021)
15. Bastián Saavedra Cornejo. M Sc. Eng. mention in Industrial Engineering (2021) (work conjoint with G. Lagos)
14. Luis Rojo-González. M. in Statistics and Operations Research [Universitat Politècnica de Catalunya (UPC), Spain] (2021) (work conjoint with M. Pérez Casany)
13. Javiera García-Leal. M Sc. Eng. mention in Industrial Engineering (2020) (work coinjoint with A.T. Espinoza Pérez)
12. Matías Anabalón. M Sc. Eng. mention in Industrial Engineering (2020)
11. Natalia Jorquera-Bravol. M Sc. Eng. mention in Industrial Engineering (2020) (work coinjoint with A.T. Espinoza Pérez)
10. Manuel Villalobos-Cid. Postdoct (2020-2022)
9. Luis Osorio-Valenzuela. Postdoct (2019)
8. Joaquín Maldonado. M Sc. Eng. mention in Industrial Engineering (2019)
7. Luis Rojo-González. M Sc. Eng. mention in Industrial Engineering (2019)
6. Nelson Troncoso. M Sc. Eng. mention in Industrial Engineering (2019)
5. Patricio J. Araya-Córdova. M Sc. Eng. mention in Industrial Engineering (2017)
4. Fabián Díaz-Núñez. M Sc. Eng. mention in Industrial Engineering (2017) (work conjoint with J. Pereira)
3. Denisse Ávalos. M Sc. Eng. mention in Industrial Engineering (2016)
2. Franco Quezada. M Sc. Eng. mention in Industrial Engineering (2016)
1. Manuela Vera. M Sc. Eng. mention in Industrial Engineering (2016)